

The first constellations were simple patterns of stars, picked out by imaginative humans thousands of years ago. A good knowledge of the sky had practical uses. Bright stars and constellations were navigation aids for travelers at night, their risings and settings provided a simple clock, and their annual progression around the sky acted as a calendar. The sky also became a picture book in

PATTERNS IN **THE SKY**

which storytellers could imagine the starry outlines of gods, heroes, and mythical beasts. All civilizations had their own constellations, based on their own culture. Those we use today stem from a group of 48 known to the ancient Greeks around 2,000 years ago. These were supplemented by others invented by astronomers in the 16th to 18th centuries, particularly in the far southern sky, which the Greeks could not see. In the 1920s, the International Astronomical Union, astronomy's governing body, officially recognized a total of 88 constellations that fill the sky from pole to pole with no gaps between them. Although constellations have outgrown their original purpose in this age of computer-controlled telescopes on Earth and in space, they still serve as a useful way of identifying the general area of sky in which a celestial object lies. They also provide a connection with the original stargazers who first looked at the sky and tried to understand the Universe around them.

◀ **Star trails**

A whirling pattern of lights is seen in the sky above the Atacama Large Millimeter/submillimeter Array (ALMA) in Chile. The streaks are star trails captured by a long-exposure photograph. Although the stars appear to circle the southern celestial pole, the movement is really that of the Earth rotating on its axis.